



Final Project - I2C Bus Communication

Group 2

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Embedded Systems III

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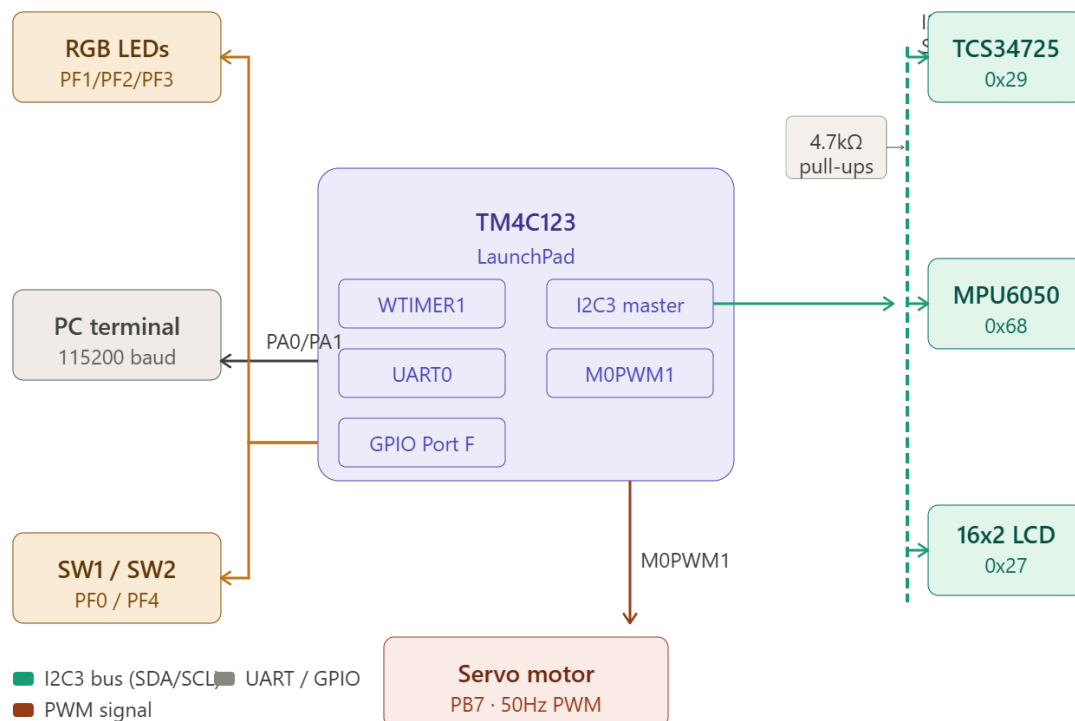
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1. Introduction

The objective of this project is to design and implement a multi-device I2C-based embedded system that integrates several sensors on a shared communication bus. The system receives color and motion data from an MPU6050 IMU and a TCS34725 color sensor. The information is processed and then displayed on an LCD and controls a servo in real time. The system focuses on firmware-level system integration, including I2C bus management, sensor data acquisition and processing, as well as timing constraints, without the use of wireless communication. From this project, students will learn essential embedded systems concepts, including: multi-device bus management, sensor data processing, real-time constraints, and system integration.

2. System Design and Architecture

Block Diagram of the System

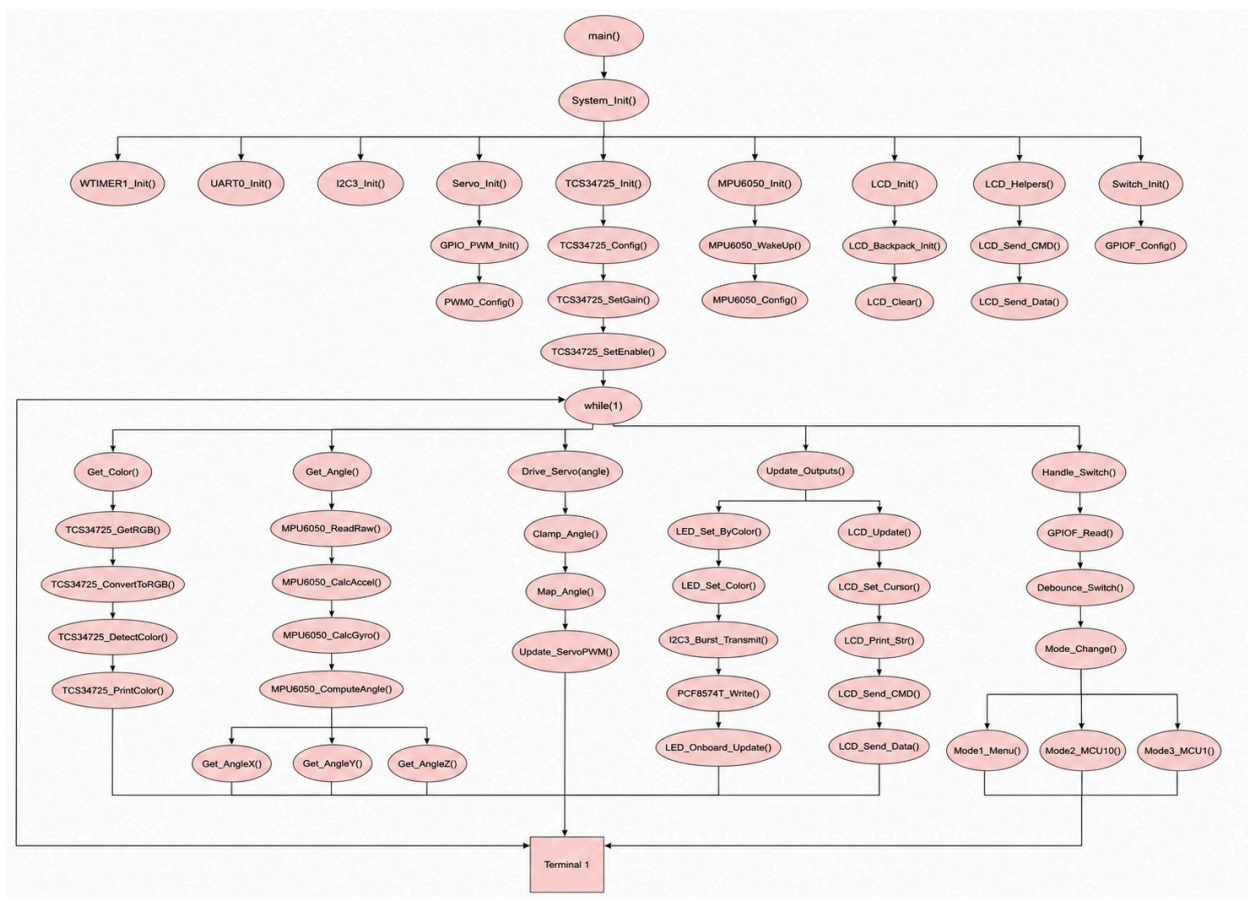


System Components

- TM4C123 LaunchPad
 - WTIMER
 - UART0
 - I2C3
 - PWM
- TCS34725 RGB Color Sensor
- MPU6050 6-DOF IMU
- 16x2 LCD Display with I2C Backpack
- Angular Servo Motor
- 4.7kΩ Pull-Up Resistors
- On-board RGB LEDs
- On-board SW1 and SW2
- Serial Terminal

Hardware-Software Interaction

Call Graph



Functional Description of Each Module

Module	Description
WTIMER	WTIMER1 is initialized for a 16MHz system clock divided for an accurate 1 millisecond resolution. It is configured as a 32-bit hardware timer that generates 1 millisecond timing delays through DELAY_1MS() function. The module is used throughout the system for timing-critical operations including sensor sampling intervals, LCD and UART update timing, and servo motion delays.
Servo	Generates a 50Hz hardware PWM signal to control servo angle. Specified angles are mapped to pulse widths between 1.0 ms and 2.0ms for angles within the range -90 to 90 degrees. The system updates the PWM duty cycle in real-time using IMU angle data to control servo movement.
TCS34725	Initializes the TCS34725 RGB color sensor at I2C address 0x29 via I2C3. The sensor measures 16-bit red, green, blue, and clear light intensity values which are normalized to a 0-255 range by dividing each channel by the clear channel value. The dominant color is determined by comparing normalized R, G, B values and the result drives the onboard LED indicators and LCD color display. Integration time is set to 2.4ms with 1x gain.
MPU6050	Initializes the MPU6050 6-DOF IMU sensor at I2C address 0x68 via I2C3. The accelerometer is configured for $\pm 2g$ range (16,384 LSB/g) and the gyroscope for $\pm 250^\circ/s$ range (131 LSB/ $^\circ/s$). Raw 16-bit signed accelerometer and gyroscope data are read via burst I2C reads and processed into physical units. Tilt angles ArX, ArY, and ArZ are computed using atan2() on the processed accelerometer data and converted from radians to degrees. The pitch angle (ArY) is used to drive the servo motor in real-time.
16x2 LCD Display	Drives a 16x2 HD44780 LCD through a PCF8574T I2C port expander at address 0x27 via I2C3. The display is initialized using the standard 4-bit mode power-on sequence with EN pulse toggling transmitted as burst I2C writes.
Switch	Two onboard pushbutton switches on Port F are configured with internal pull-up resistors and edge-triggered interrupts via GPIOPortF_Handler(). SW1 (PF4) toggles the system between active and stopped states by setting SW1_Flag in the ISR which main() polls each cycle. SW2 (PF0) sets SW2_Flag to acknowledge a display update request.

LED	Three onboard RGB LEDs on Port F are configured as GPIO outputs. PF1 (Red), PF2 (Blue), and PF3 (Green) are driven based on the color detected by the TCS34725 sensor. The corresponding LED lights when its color is dominant and all LEDs are turned off when no dominant color is detected or the system is stopped via SW1.
UART0	Initializes UART0 at 115,200 baud (IBRD=8, FBRD=44) with 8 data bits, 1 stop bit, no parity, and FIFOs enabled on PA0 (Rx) and PA1 (Tx). Provides OutChar, OutString, SendInteger, and SendFloat helper functions. During operation the system transmits formatted sensor data to the PC terminal every 1 second in the format: "Color: R=X, G=X, B=X"
I2C3	Initializes I2C3 as master at 100kHz Standard Mode using pins PD0 (SCL) and PD1 (SDA) with external 4.7k Ω pull-up resistors to 3.3V. The TPR register is set to 7 for 100kHz operation at 16MHz system clock. Provides single-byte transmit, single-byte receive, burst transmit, and burst receive functions with error detection. All three I2C slave devices — TCS34725 (0x29), MPU6050 (0x68), and LCD backpack (0x27) — share this single bus.

3. Development Steps and Task Assignments

Project Planning and Milestones

Phase	Description	Dates
Step 1 — Design	Requirements review, system architecture, pin assignments	Apr 20–22
Step 2 — Modules	Individual module implementation and unit testing	Apr 23 – May 4
Step 3 — Integration	Full system integration, debugging, final submission	May 5–13

Development Steps

1. Circuit design — identified pinouts, wired I2C bus with pull-ups, connected servo and LCD
2. WTIMER1 — implemented and verified 1 ms delay accuracy on oscilloscope ($\pm 5\%$)
3. UART0 — implemented serial output at 115,200 baud; verified on PC terminal
4. I2C3 — implemented master driver with single-byte and burst transactions; verified ACK bits

5. TCS34725 — verified ID register, configured sensor, tested RED/GREEN/BLUE detection
6. MPU6050 — verified WHO_AM_I register, confirmed Z-axis reads ~16,384 LSB at rest
7. LCD — implemented 4-bit init sequence, verified text output and startup name sequence
8. Servo — implemented PWM at 50 Hz, verified startup motion sequence, corrected pulse range to 1.0–2.0 ms
9. Integration — wrote main.c combining all modules with 100 ms sensor loop and 1s display update.

Testing and Debugging Procedures

1. Verified servo 50Hz PWM frequency by measuring PB7 signal on oscilloscope confirming 20ms period.
2. Identified and resolved I2C bus corruption by testing each component individually both physically and on software.
3. Tested all module tests individually to confirm each module was working before implementing the full system integration.
4. Debugged LCD conflict with servo — resolved by initializing LCD before servo and reinitializing I2C3 after servo
5. Debugged servo conflict by switching out the pwm because PB7 was a special pin.
6. Debugged frozen Ax=Ay=Az=3.26 values — identified individual I2C reads returning same register repeatedly, fixed by switching to burst reads using I2C3_Burst_Recieve.
7. Debugged frozen R=255 G=255 B=255 values — identified as I2C bus corruption from servo init, resolved by reordering System_Init()

Task Assignment Table/Progress Tracking

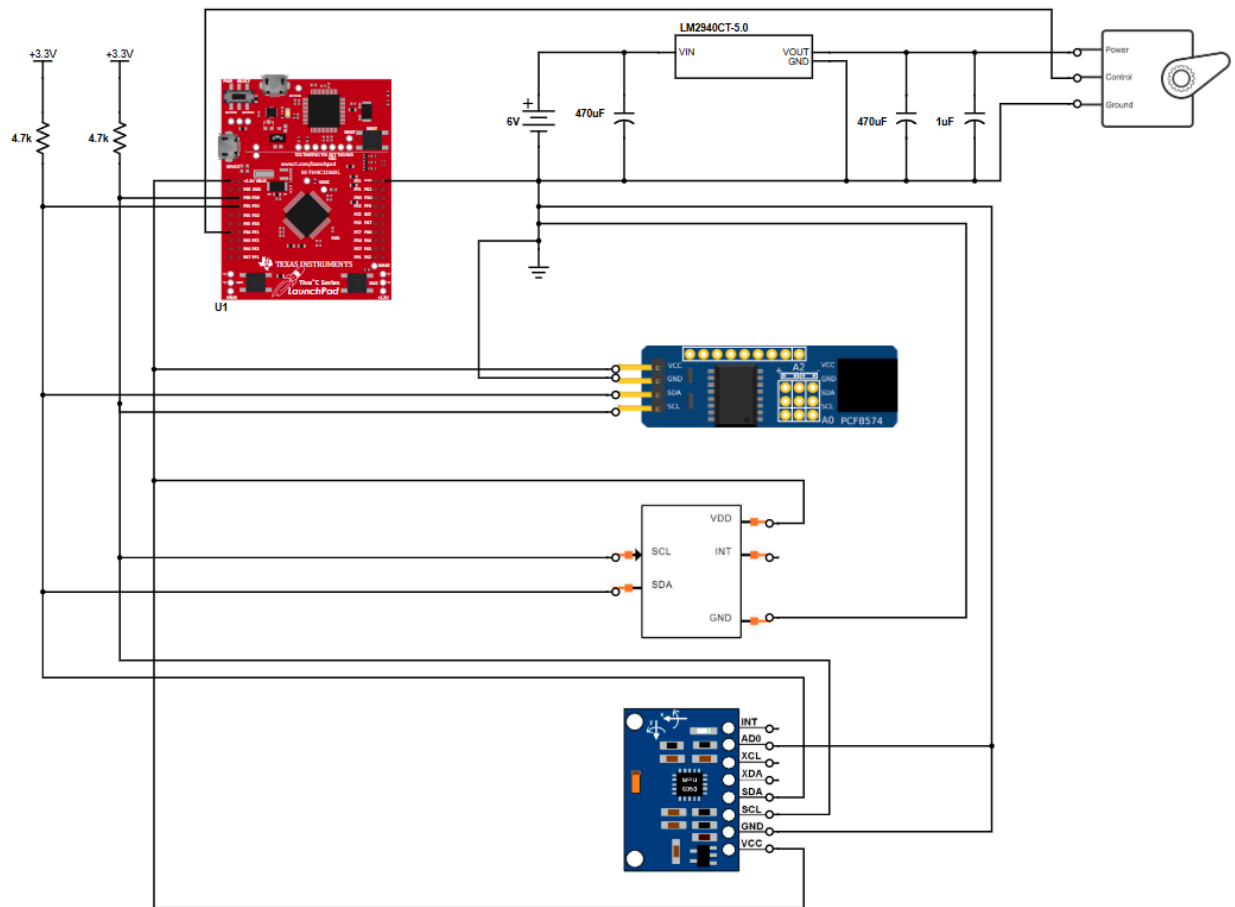
Team Member	Role/Responsibility	Assigned Tasks	Completion Date
Ana Arante	Hardware Lead	Servo PWM, ButtonLED, circuit wiring, system integration	5/13/2026
Emily Hsu	Firmware Lead	WTIMER1, UART0, I2C3 driver, MPU6050, main.c	5/13/2026
Suphia Sidiqi	Sensor Lead	TCS34725, LCD driver, startup sequence, report	5/13/2026

Collaboration Tools:

- GitHub Classroom was used to collaborate on all source files, add module test evidence, and log AI usage.
- Tera Term / PuTTY — serial terminal for UART debug output
- Oscilloscope — timing verification for I2C, PWM, and WTIMER

4. Hardware Design

Schematic Diagram



Circuit Explanation (Components and Connections)

Component	Interface	Address / Pin	Purpose
TM4C123 LaunchPad	—	Main processor	Host microcontroller
TCS34725 Color Sensor	I2C3	0x29	RGB color detection
MPU6050 IMU	I2C3	0x68	Accelerometer / gyroscope
16x2 LCD (PCF8574T)	I2C3	0x27	Text display output
Angular Servo Motor	M0PWM1 (PB7)	50 Hz PWM	IMU-driven actuation
UART0	PA0 / PA1	115200 baud	PC serial debug terminal
WTIMER1	Internal	1 ms resolution	System timing delays
Pull-up Resistors	SDA / SCL	4.7 kΩ to 3.3 V	I2C bus termination

Pin Assignments

Pin	Signal	Module	Notes
PA0	U0Rx	UART0	Receive from PC
PA1	U0Tx	UART0	Transmit to PC
PB7	M0PWM1	Servo PWM	50 Hz, 1.0–2.0 ms pulse
PD0	I2C3 SCL	I2C3 master	Clock — 100 kHz
PD1	I2C3 SDA	I2C3 master	Data — open drain
PF1	Red LED	ButtonLED	Color indicator
PF2	Blue LED	ButtonLED	Color indicator
PF3	Green LED	ButtonLED	Color indicator
PF0	SW2	ButtonLED	Display update trigger

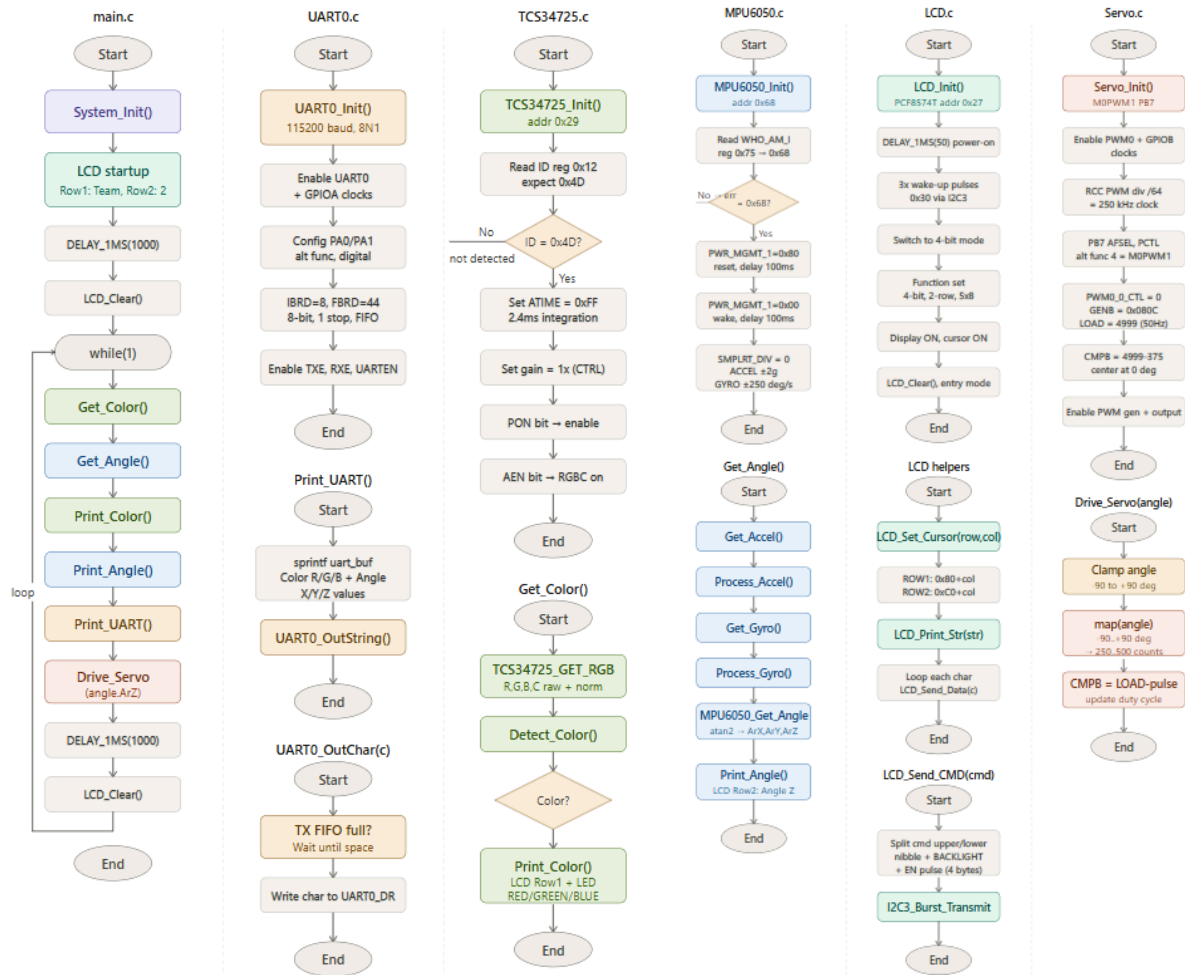
PF4	SW1	ButtonLED	Start / stop detection
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Bill of Materials (BOM)

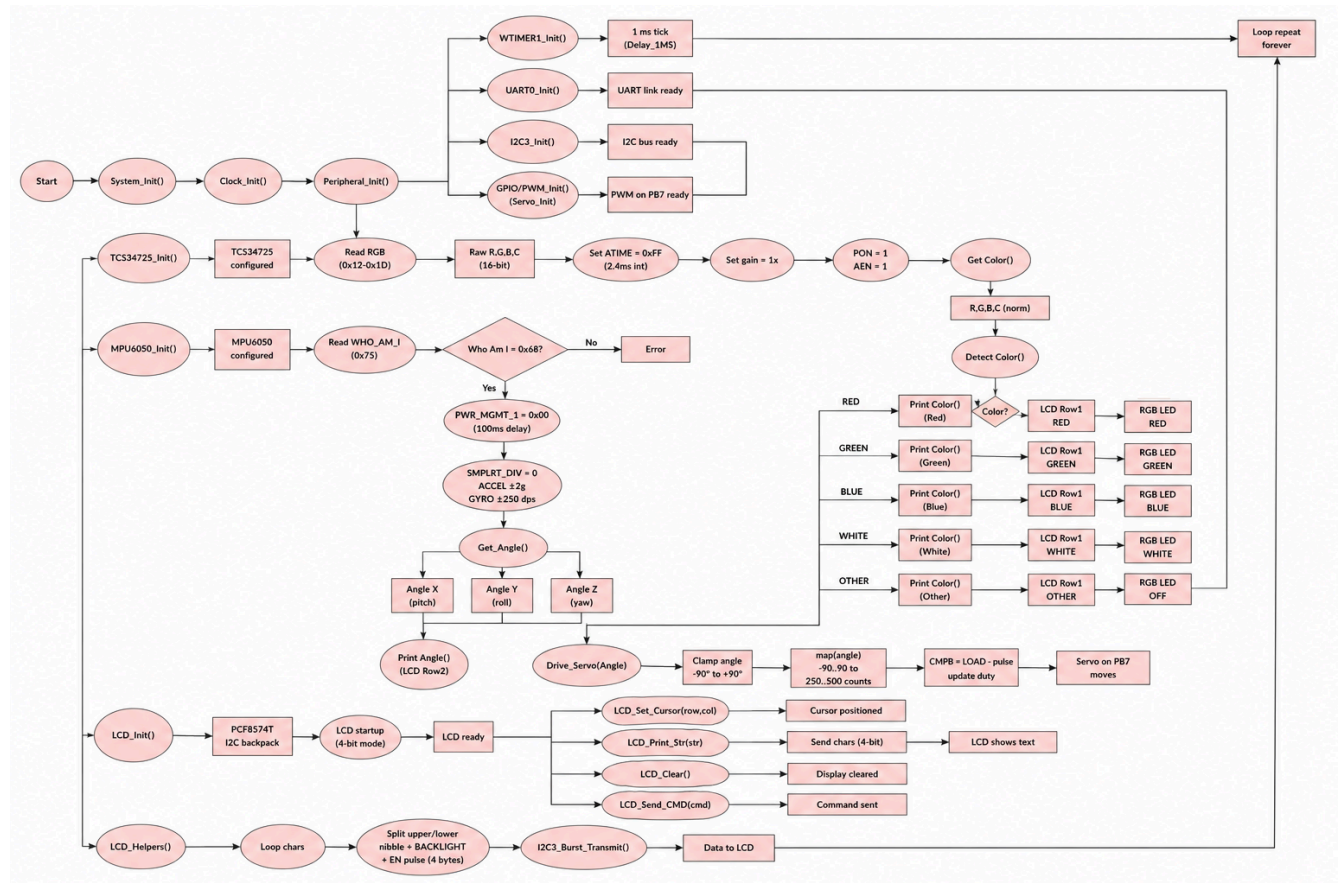
Item #	Component Name	Description	Quantity	Notes
1	TM4C LaunchPad	TM4C123 Microcontroller	1	Main Processor
2	TCS34725	RGB Color Sensor	1	Color detection I2C interface, address 0x29
3	MPU6050 IMU	Tilt/Acceleration Measurement	1	I2C interface, Address 0x68
4	Angular Servo Motor	Mechanical Actuator	1	Standard 5V, 50Hz PWM control
5	Pull-up Resistors	I2C bus termination (SDA, SCL)	2	4.7k Ω
6	PC/Laptop	Personal Computer	1	Running serial terminal for UART output
7	Serial Terminal	Tera Term	1	PC-based debugging interface

5. Software Design

Software Flowchart



Pseudocode / Data Flow Graph



Key Functions and Modules

Module 1 — WTIMER1:

Configures WTIMER1 as a 32-bit count-down periodic timer with a reload value of 15,999 counts at 16 MHz (no prescaler) to produce exactly 1 ms per tick. DELAY_1MS(ms) polls the timeout flag in a loop for precise blocking delays used by all other modules.

Module 2 — UART0:

Initializes UART0 at 115,200 baud (IBRD=8, FBRD=44) with 8 data bits, 1 stop bit, no parity, FIFOs enabled. Provides OutChar, OutString, SendInteger, and SendFloat helpers. Outputs sensor data to a PC terminal at 1 Hz.

Module 3 — I2C3:

Configures I2C3 master at 100 kHz Standard Mode (TPR=7 for 16 MHz clock). Provides single-byte and burst read/write functions with error detection. External 4.7 kΩ pull-up resistors on SDA and SCL are required.

Module 4 — TCS34725:

Verifies sensor via ID register (0x12 → 0x4D). Configures integration time to 2.4 ms and gain to 1x. Reads 16-bit raw R, G, B, and Clear values and normalizes to 0–255 by dividing by the clear channel. Detect_Color() returns the dominant color enum.

Module 5 — MPU6050:

Verifies sensor via WHO_AM_I register (0x75 → 0x68). Configures accelerometer to $\pm 2g$ (16,384 LSB/g) and gyroscope to $\pm 250^\circ/s$ (131 LSB/ $^\circ/s$). Reads 6-byte raw data for each axis and computes tilt angles using atan2.

Module 6 — LCD:

Drives the 16x2 HD44780 LCD through the PCF8574T I2C port expander at address 0x27. Implements 4-bit initialization sequence. LCD_Print_Str, LCD_Set_Cursor, and LCD_Clear provide the display interface. Startup shows team name sequence before entering main loop display.

Module 7 — Servo:

Configures PWM Module 0 Generator 0 Output B (M0PWM1) on PB7 at 50 Hz using a /64 clock divider (250 kHz PWM clock, 5000-count period). Drive_Servo maps angles from -90° to $+90^\circ$ linearly to pulse widths of 1.0–2.0 ms (counts 250–500).

6. Demonstration

Working Procedure

1. Open the UART terminal at 115200 baud. Show the terminal output confirming each module initialized successfully: "TCS34725 has been Detected", "MPU6050 detected", and LCD initialization messages.
2. The LCD displays Team on row 1 and 2 on row 2 to demonstrate two names on each row of the LCD.
3. **Color Detection (RED)** Hold a red object in front of the TCS34725 sensor. Show the LCD Row 1 updating to "Color: RED", the red onboard LED (PF1) lighting up, and the UART terminal printing the corresponding R value higher than G and B.
4. **Color Detection (GREEN)** Replace the red object with a green object. Show the LCD updating to "Color: GREEN", the green LED (PF3) lighting up, and the UART values reflecting a dominant G channel.
5. **Color Detection (BLUE)** Replace with a blue object. Show the LCD updating to "Color: BLUE", the blue LED (PF2) lighting up, and the UART values reflecting a dominant B channel.
6. **IMU Tilt Detection** Hold the board flat and show the UART printing the current X, Y, Z angles. Then tilt the board forward and backward along the Y axis and show the angle values changing in real time on both the UART terminal and the LCD Row 2.

7. **Servo Motor Response** With the board flat show the servo at its center position (0 degrees). Tilt the board toward -90 degrees and show the servo rotating to one extreme. Tilt toward +90 degrees and show the servo rotating to the other extreme. Show the servo following the tilt smoothly in real time.
8. **SW1 Start/Stop** Press SW1 and show the system stopping — LEDs turn off, servo stops updating. Press SW1 again and show the system resuming normal operation.
9. **UART Output** Point the camera at the terminal and show the formatted output updating every 1 second in the correct spec format: **Color: R=XXXXX, G=XXXXX, B=XXXXX | Angle: X=X, Y=X, Z=X**
10. **Full System Integration** Show all outputs simultaneously — LCD displaying color and angle, LED lit for detected color, servo moving with board tilt, and UART terminal streaming data — demonstrating all 7 modules working together as one integrated system.

Observed Results

- TCS34725 correctly identifies RED, GREEN, and BLUE objects with corresponding LED lighting
- MPU6050 Z-axis reads approximately 16,384 LSB at rest confirming 1g gravity reading
- Servo moves smoothly through -90° to +90° range with no jitter
- LCD displays color names and angles updating at 1 Hz
- UART terminal shows properly formatted sensor data at 1 Hz
- I2C clock confirmed at 100 kHz on oscilloscope with ACK bits present for all three devices
- PWM frequency confirmed at 50 Hz with correct pulse width range.

Video Link of Demonstration:

https://youtube.com/shorts/lZr4alfZj-Q?si=KCIB--A5N_9JK_0g

8. Conclusion

The team successfully designed and implemented a fully integrated I2C sensor fusion system on the TM4C123 LaunchPad. All seven modules were implemented individually, unit tested, and integrated into a cohesive real-time system. The system reliably reads color data from the TCS34725, computes tilt angles from the MPU6050, displays results on the LCD and UART terminal, and controls a servo motor in real time, all sharing a single I2C bus at 100 kHz. Some complications we were running into involved the servo not properly rotating when the gyroscope was turning. In addition, the gyroscope values and the color sensor values were not updating. We fixed this issue by double checking our pins to check for special pins or pins that need unlocking. The pin used on the servo was a special pin, so

we changed our pwm1 to pwm2 which uses PB4. This problem was the reason our integrated system was falling through, with this fix we were able to successfully implement our system.

9. References

- [1] Texas Instruments, TM4C123GH6PM Microcontroller Datasheet, Rev. 2016.
- [2] ams AG, TCS34725 Color Sensor Datasheet, Rev. 1.0.
- [3] InvenSense, MPU-6000/MPU-6050 Register Map and Descriptions, Rev. 4.2.
- [4] J. Huynh and Dr. M. He, CECS 447 Final Project: I2C Bus Communication System, Cal State Long Beach, 2026.